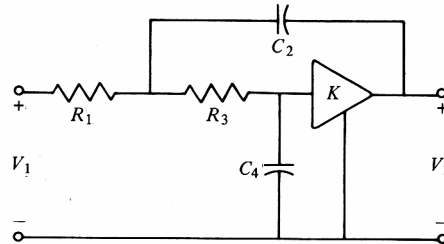


Hardware Project No. 2: Active Low-Pass Filter

For this project, you will design and build an active low-pass filter. These have multiple uses in biosignal acquisition, tailoring the measurement to the frequency characteristics of the signal under consideration, as well as performing an anti-aliasing function. The figure below shows a Sallen-Key type low-pass filter design.



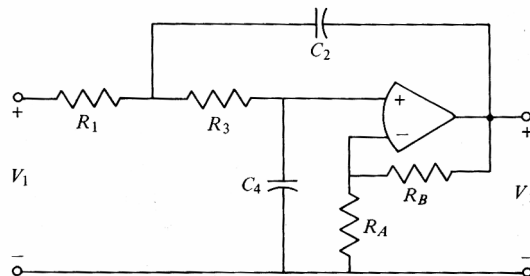
The Sallen-Key design uses two resistors, two capacitors and an amplifier. The design parameters of this filter are the cutoff frequency ω_n , the DC gain H_0 and the filter Q , and are given by:

$$\omega_n = \frac{1}{\sqrt{R_1 R_3 C_2 C_4}}$$

$$\frac{1}{Q} = \sqrt{\frac{R_3 C_4}{R_1 C_2}} + \sqrt{\frac{R_1 C_4}{R_3 C_2}} + (1 - K) \sqrt{\frac{R_1 C_2}{R_3 C_4}}$$

$$H_0 = K$$

This gives us three equations with five unknowns (four circuit components, and the gain), which leaves us with some design choices to make for a filter; welcome to being a Boy or Girl Engineer! A common set of constraints to impose is to have equal-valued capacitors and a gain (K) of 2. This configuration using an op-amp is the following.



This circuit should explain for the decision for using a gain of 2.

You will design and build a Butterworth low-pass filter using equal capacitors (pick a capacitance...your choice!) and a gain of 2. I leave the cutoff frequency as a parameter of

your particular design. Let the signal that you would choose to acquire with this amplifier guide you: of the bioelectric signals we will be discussing this semester, surface EMG should be 500 Hz or less, fine-wire should be 1 kHz or less, EKG should be about 150 Hz, EEG should be around 50 Hz. Also, be advised that this would be the anti-aliasing filter in a system.

If you are scared by having so much freedom, then use 500 Hz as your cutoff frequency (for a very low corner frequency, you will be unable to make many measurements in the passband; 500 Hz will make accurate corner frequency and passband measurements relatively easy). Now all you need is the filter Q, and it turns out that you have that from the above specifications; if you're not sure, then you need to do a little research (hint: it's a *Butterworth* filter). The Internet can be a fine source of info, as can the musty archives of The Cooper Union Library—you may even find books on filter design there! Wow! On the basis of the above, derive what the filter parameter equations simplify to; your report should include this information. The simplify nicely, and tell you how to compute your resistor values once you've picked a capacitor.

A hint: pick (and match *very* closely) your capacitors first. Resistors are always easy to tweak by making series and parallel combinations, but capacitors are much trickier. You will note that getting your component values right is sort of important to getting the corner frequency and response that you design for.

As for the testing, you will need to provide the frequency response of the filter; be sure to measure lots of points around the corner frequency, and make some measurements to show that the stopband falls off at the rate it should, i.e., 40 dB/decade (12 dB/octave) for a second-order filter such as this one. And remember, that rate of falloff occurs once one gets some distance from the corner frequency. So, if you design for a cutoff of 500 Hz, and then measure the rolloff over an octave from 500 Hz to 1 kHz, or a decade from 500 Hz to 5 kHz, you will get lousy numbers. Better to look at an octave from, say 1 kHz to 2kHz. Or, 2 kHz to 4 kHz. You may notice that at some point, things start to level off and you start getting a constant stopband gain; welcome to the world of real rather than ideal op amps, where gain-bandwidth product is a real thing.

Write a report as per the previous hardware assignment.

Due Date

March 12, 2026

We'll demo circuits in the lab a little. It's always fun to play "spin the frequency dial" for a little bit and watch sine waves get bigger and smaller. You can hand me the report then, or a few days afterwards. But you probably don't want to be writing it over spring break.